

1. Administrative & Study Identification

Full Study Title: KEYMAKER-U01 Substudy 2: A Phase 2, Umbrella Study With Rolling Arms of Investigational Agents in Combination With Pembrolizumab in Treatment Naïve Patients With PD-L1 Positive Advanced Non-small Cell Lung Cancer (NSCLC) **Short Title/Acronym:** KEYMAKER-U01B / MK-3475-01B **Protocol Number:** MK-3475-01B **NCT Number:** NCT04165083 **Posting ID:** 718645122135314554 **Trial Status:** Active but not currently recruiting (ACTIVE_NOT_RECRUITING) **Sponsor / Company Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Pembrolizumab (MK-3475, Trade Name: Keytruda) in combination with investigational agents (including MK-4830 and MK-0482) **Phase of Development:** Phase II **Medical Monitor:** Merck Sharp & Dohme LLC
Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of pembrolizumab (MK-3475) in combination with MK-4830 (or other investigational agents such as MK-0482, which was added with Amendment 6) in treatment-naïve participants with advanced squamous or non-squamous NSCLC that is PD-L1 positive. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS) and Objective Response Rate (ORR) per RECIST 1.1, and to continuously monitor safety, tolerability, and study treatment discontinuations due to adverse events.

2.2 Background & Rationale While PD-1/PD-L1 inhibitors have revolutionized the treatment of NSCLC, resistance and tumor microenvironment immunosuppression remain significant challenges. Combining pembrolizumab with novel investigational agents aims to prevent tumor-induced effector T-cell dysfunction. This study is one of three pembrolizumab substudies being conducted under one pembrolizumab umbrella master protocol (MK-3475-U01/KEYMAKER-U01). The master screening protocol is MK-3475-U01(KEYMAKER-U01) - NCT04165798.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Master Protocol Substudy) **Control Type:** Active Comparator (Rolling arms comparing combinations vs. standard expected outcomes) **Blinding:** Open-label **Randomization:** Sequential/Non-randomized (Rolling arms assignment)

3.2 Planned Enrollment & Duration Targeted Enrollment: 120 participants **Number of Sites:** International (United States, Hungary, Israel, Italy, Korea, Poland, Spain) **Estimated Study Start:** 18-Dec-2019 **Estimated Primary Completion:** 30-Oct-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has a histologically- or cytologically-confirmed diagnosis of Stage IV squamous or non-squamous NSCLC. 2. Has non-squamous NSCLC and is not eligible for an approved targeted therapy. 3. Is able to provide an archival tumor tissue sample (collected within 5 years or within the interval from completion of last treatment but before screening) or a newly obtained core/excisional biopsy of a non-irradiated tumor lesion obtained within 90 days of treatment initiation. 4. Has not received prior systemic treatment for metastatic NSCLC. 5. Has a programmed cell death ligand 1 (PD-L1) tumor proportion score (TPS) $\geq 1\%$. 6. Is able to complete all screening procedures within the 35-day screening window. 7. Has adequate organ function within 10 days of initiation of study treatment. 8. Male participants must agree to use contraception and refrain from donating sperm during the treatment period and for at least 120 days after the last dose of study treatment. 9. Female participants must not be pregnant or breastfeeding, and must either not be of childbearing potential (WOCBP), or if WOCBP, agree to use contraception during treatment and for at least 120 days post-last dose.

4.2 Exclusion Criteria 1. Has a diagnosis of small cell lung cancer (SCLC). 2. Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy (>10 mg daily prednisone equivalent) or any other immunosuppressive therapy within 7 days before the first dose. 3. Has a known additional malignancy that is progressing or has required active treatment within the past 2 years. 4. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 5. Has an active autoimmune disease requiring systemic treatment in the past 2 years. 6. Has a history of (noninfectious) pneumonitis that required steroids or current pneumonitis. 7. Has an active infection requiring systemic therapy or a known history of HIV, Hepatitis B, or active Hepatitis C virus infection. 8. Has clinically significant cardiac disease, including unstable angina, acute myocardial infarction within 6 months of Day 1, or NYHA Class III or IV congestive heart failure. 9. Has known psychiatric or substance abuse disorders interfering with study cooperation. 10. Has had major surgery <3 weeks before the first dose. 11. Has received prior radiation therapy to the lung (>30 Gy) within 6 months of the first dose. 12. Has received any prior immunotherapy and discontinued due to a severe or worse immune-related adverse event (irAE). 13. Has had chemotherapy or biological cancer therapy within 4 weeks before the first dose or has not recovered to CTCAE Grade 1 or better from prior

therapies. 14. Has received a live vaccine within 30 days before the first dose. Licensed COVID-19 vaccines (mRNA, adenoviral, or inactivated) are permitted; investigational vaccines are not allowed. 15. Has received prior systemic cytotoxic chemotherapy, other targeted/biological antineoplastic therapy for metastatic disease, or prior therapy with an anti-PD-1, anti-PD-L1, anti-PD-L2 agent, or other immuno-regulatory targeting therapy. 16. Is currently participating or has participated in an investigational agent/device study within 4 weeks before the first dose. 17. Has a severe hypersensitivity reaction to monoclonal antibodies (including pembrolizumab) or excipients. 18. Is pregnant, breastfeeding, or expecting to conceive/father children starting with screening through 120 days post-last dose. 19. Has had an allogenic tissue/solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pembrolizumab (Keytruda, ATC: L01FF02) combined with investigational agents MK-4830 or MK-0482. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Until documented disease progression, unacceptable toxicity, or investigator decision.

5.2 Concomitant & Prohibited Therapy Permitted: Supportive care medications as defined per standard oncology guidelines; licensed COVID-19 mRNA, adenoviral, or inactivated vaccines. **Prohibited:** Systemic immunosuppressive therapies (e.g., corticosteroids >10 mg/day prednisone equivalent within 7 days of dose), live vaccines within 30 days, investigational vaccines, and concurrent investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -35 to 0	Informed Consent, Archival/New Tumor Biopsy (collected <5 years or <90 days respectively), Imaging, Labs
Baseline	Day 0	Final Eligibility Confirmation (organ function <10 days prior), First Dose
Treatment Cycles	Q3W / Q6W	Pembrolizumab + Investigational Agent Infusion, Safety Labs, AE Assessment
End of Treatment	+ 30 Days	Final Vital Signs, Exit Interview, Safety Follow-up
Follow-up	Up to 120 Days	Survival Tracking, Pregnancy/Contraception monitoring (120 days post-last dose)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR). The percentage of participants in the analysis population who had a Complete Response (CR: disappearance of all target lesions) or

a Partial Response (PR: $\geq 30\%$ decrease in the sum of diameters of target lesions). **Measurement Method:** Radiological assessment per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1).

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) According to RECIST 1.1. * Number of Participants Who Experience One or More Adverse Events (AEs). * Number of Participants Who Discontinued Study Treatment Due to an Adverse Event (AE).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of any unfavorable and unintended signs, symptoms, or disease worsening. **Serious Adverse Events (SAEs):** Reported within 24 hours to the Sponsor/IRB. **Data Monitoring Committee (DMC):** Utilized for the overarching KEYMAKER-U01 umbrella master protocol to evaluate ongoing safety and tolerability of rolling arms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy endpoints; Safety Analysis Set for patients who received ≥ 1 dose of study treatment. **Sample Size Justification:** Evaluated per individual rolling arm, targeting an overall $N = 120$ to detect meaningful clinical responses above historical controls. **Handling of Missing Data:** Participants with missing assessments are generally censored at the date of the last valid assessment for time-to-event endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard clinical oncology monitoring. **Audit Trail:** Secure EDC (Electronic Data Capture) systems with independent central review for RECIST 1.1 assessments. **EMA Regulatory Source:** [Keytruda EPAR Product Information](#)

11. Study Identifiers & Governance

Primary ID: MK-3475-01B **Registry Number:** NCT04165083 **Posting ID:** 718645122135314554 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC (Merck) **Study Title:** KEYMAKER-U01 Substudy 2: A Phase 2, Umbrella Study With Rolling Arms of Investigational Agents in Combination With Pembrolizumab in Treatment Naïve Patients With PD-L1 Positive Advanced Non-small Cell Lung Cancer (NSCLC)

Official Regulatory Title: KEYMAKER-U01 Substudy 2: Efficacy and Safety Study of Pembrolizumab (MK-3475) When Used With Investigational Agents in Treatment-naïve Participants With Anti-programmed Cell Death Receptor Ligand 1 (PD-L1) Positive Advanced Non-small Cell Lung Cancer (NSCLC) (MK-3475-01B/KEYMAKER-U01B)

12. Clinical Framework & Ontology

Primary Condition / Disease Area: Carcinoma, Non-Small-Cell Lung
Semantic Type: Neoplastic Process **Disease Definition:** A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, squamous epithelium are called squamous cell carcinomas, and transitional epithelium are called transitional cell carcinomas. **Clinical Ontologies / Codes:** * MeSH Code: D002277 * SNOMED ID: 1187425009 * HPO Code: HP:0030731 * ATC Code (Pembrolizumab): L01FF02
Preferred UMLS Name (Drug Class): Programmed Cell Death 1 Receptor Inhibitors **Diagnosis Threshold:** Stage IV, PD-L1 TPS ≥ 1%, Treatment-Naïve

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for advanced NSCLC management **Education Protocol (HCP):** Investigational product administration, immune-related AE (irAE) management training **Education Protocol (Patient):** Informed consent, contraception compliance (120-day rule), and side-effect reporting **Quality Audit Mechanism:** Clinical site monitoring and central blinded review of radiological imaging (RECIST 1.1)

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression, unacceptable toxicity, or trial completion **Onsite Visit Cadence:** Aligned with infusion cycles (typically Q3W) **Remote Monitoring Frequency:** Between treatment cycles for safety evaluation **Checklist Review Interval:** Prior to every IV infusion for AE and concomitant medication review

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				